

Math 241, F12
Quiz 13

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Evaluate the line integral $\int_C x \sin y \, ds$ where C is the line segment from $(0, 3)$ to $(4, 6)$.

Solution: A parametric equation for the line segment from $(0, 3)$ to $(4, 6)$

$$\gamma(t) = (1-t)\langle 0, 3 \rangle + t\langle 4, 6 \rangle = \langle 4t, 3+3t \rangle, 0 \leq t \leq 1$$

$$\Rightarrow x(t) = 4t, y(t) = 3+3t, 0 \leq t \leq 1$$

$$\begin{aligned} \int_C x \sin y \, ds &= \int_0^1 (4t) \sin(3t+3) \cdot \sqrt{4^2 + 3^2} \, dt \\ &\quad \rightarrow = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \\ &= 20 \int_0^1 t \sin(3t+3) \, dt \\ &= 20 \left[t \left(-\frac{\cos(3t+3)}{3} \right) \Big|_{t=0}^{t=1} - \int_0^1 \left(-\frac{\cos(3t+3)}{3} \right) dt \right] \\ &= 20 \left[-\frac{\cos 6}{3} + \frac{1}{9} (\sin 6 - \sin 3) \right] \end{aligned}$$

Problem 2. (5 points) Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y, z) = (x+y)\mathbf{i} + (y-z)\mathbf{j} + z^2\mathbf{k}$ and C is given by the vector function $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j} + t^2\mathbf{k}$, $0 \leq t \leq 1$.

Solution: $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{r}' \, dt$

$$\begin{aligned} &= \int_0^1 [(x(t)+y(t))\mathbf{i} + (y(t)-z(t))\mathbf{j} + z^2(t)\mathbf{k}] \cdot [2t\mathbf{i} + 3t^2\mathbf{j} + 2t\mathbf{k}] \, dt \\ &= \int_0^1 \langle t^2+t^3, t^3-t^2, t^4 \rangle \cdot \langle 2t, 3t^2, 2t \rangle \, dt \\ &= \int_0^1 (t^2+t^3)(2t) + (t^3-t^2)(3t^2) + t^4(2t) \, dt \\ &= \int_0^1 (5t^5 - t^4 + 2t^3) \, dt \\ &= \left[\frac{5}{6}t^6 - \frac{1}{5}t^5 + \frac{1}{2}t^4 \right]_{t=0}^{t=1} \\ &= \frac{17}{15} \end{aligned}$$